

Exploring a Digital Art Collection through Drawing Interactions with a Deep Generative Model

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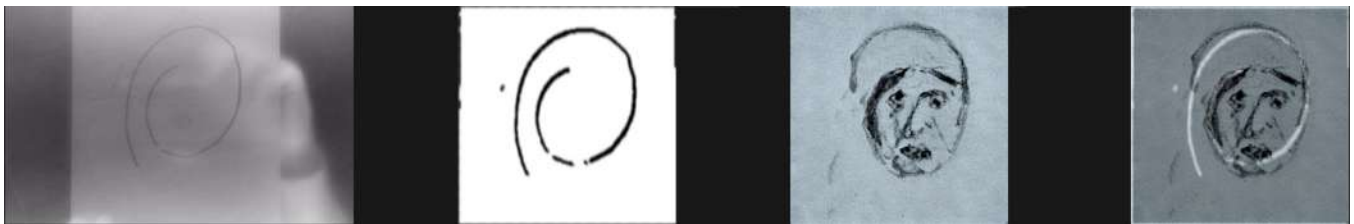


Figure 1: Inferring a Munch-like sketch from a hand-drawn line. From left to right: 1) infrared webcam image from beneath the drawing surface. 2) The cleaned input image. 3) The synthesized sketch. 4) A composite image of the input image over the synthesized sketch.

ABSTRACT

New Snow is an interactive drawing table that investigates human interaction with a deep generative model based on Edvard Munch's sketching practice. Through drawings with pen and paper, the user can interact with the model which will return synthetic sketches based on the input drawings in real time. The model is a reflection of the training data, and it is thus constrained to representing images within the latent space of Edvard Munch's sketching practice. As the user familiarizes themselves with the model it allows them to become sensitized to the visual aesthetic belonging to this practice. This potential for familiarization with the aesthetic of a dataset via the model has implications for human-AI interaction and non-verbal art mediation.

CCS CONCEPTS

• **Human-centered computing** → *Interface design prototyping; Interaction design; Interaction devices*; • **Applied computing** → *Fine arts*; • **Computing methodologies** → *Machine learning*.

KEYWORDS

sketching, embodied interaction, interaction, stylegan, fine art

ACM Reference Format:

Christian Sivertsen, René Haas, Halfdan Hauch Jensen, and Anders Sundnes Løvlie. 2023. Exploring a Digital Art Collection through Drawing Interactions with a Deep Generative Model. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3583902>

1 INTRODUCTION

Museums across the world have built up large collections of digitized artwork collections. This vast amount of material can be difficult to present to an audience. Several projects investigate ways to expose museum visitors to large datasets and allow for browsing or reauthoring the content [17–19, 23]. However, collection interfaces are often designed to offer overview and searchability rather than engagement with the artworks. To support the mediation of large art collections, interfaces are necessary that support alternative and embodied modes of engagement.

Tapping into recent developments in image synthesis, New Snow attempts to explore a new way to engage with a large collection of digitized artworks through a novel machine-learning-enabled interface. Deep Generative Models (DGM) are models that can generate images that resemble the training data. These have recently become well-known through the current wave of text-to-image systems like Midjourney, DALL-E [21] and Stable Diffusion [5] that allow for anyone to generate synthetic images in the style of famous artists, that is, artists whose works have a large presence on the internet from where much of the data for the underlying datasets are found. In contrast to these systems, New Snow employs a model trained specifically on the drawings of Edvard Munch (fig. 2), to allow

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583902>



Figure 2: The upper half of the image shows samples from the original sketch data. The lower half shows synthetic samples from the StyleGAN model

museum visitors to engage with this well-known artist’s drawing practice through their own drawing actions. New Snow is an interactive drawing table that offers museum visitors an embodied mode of interaction, where the system responds to the user’s drawing by adding lines and patterns generated by the DGM. This simulates an experience of the artist “filling in” the lines drawn by the visitor.

This project aims for three main contributions. First, it enables a way for visitors to engage with a large corpus of artworks that could not feasibly be explored individually, through the proxy of a DGM offering a synthesis of the data. Second, the system allows an embodied and creative engagement through the drawing actions of the visitor, and the interplay between the visitor and the system. Third, the system explores a novel use of a DGM, in which the user’s efforts to learn how to interact with the model are offered as a way to learn about the aesthetics of Munch’s drawings. As the user investigates the model through the drawing actions, the user learns about its qualities, and by proxy certain qualities of the artworks constituting the underlying dataset. This means that building a mental model of the system becomes a way of learning about the aesthetics of the drawings, and the image synthesis becomes an enabler of the exploration rather than the end goal.

2 RELATED WORK

Large databases of cultural heritage collections are usually accessed through search interfaces letting the users find content based on written prompts and filters. This is useful for situations where the users have a good idea of what they are looking for. However, when the domain is unfamiliar to the user, curation and recommendation

are often used as a way of guiding the user to relevant content. Earlier projects have attempted different visualization strategies for large cultural heritage datasets like *T_Visionarium II* [23], *Cloud-browsing* [19] and *ECLoud WWI* [17] that all utilize large projection surfaces to display content and allow the user to browse through the individual data objects.

The two projects *Draw to Art* [11] and *Draw to Art: Shape Edition* [10] explores visual search by allowing users to draw images on a tablet surface to query a large art database for artworks matching the drawing. In the first version, the match is based on classifying the input image as a word and then returning images relating to that word. In the second version, the system returns images with shapes matching the input image, which is constrained to simple geometric shapes. This difference marks a significant change as it enables exploration that is driven by visual concepts such as composition and shapes.

Human-AI interaction research stipulates that the system should provide the user with clear concepts of its capabilities [2] by being *explainable* or *transparent* e.g. [3, 12]. Another related concept is that of *interpretable AI* [6, 8], asserting that the users of a system should be able to interpret the underlying reasons for the output.

Building expertise in the interaction with image synthesis models is seen in *prompt engineering*, the practice of developing text-based prompts through optimization or exploration that makes text-to-image or text-to-text generation systems generate the content intended by the user. According to Oppenlaender, achieving the best results requires a deep understanding of the underlying dataset [20].

Based on the works above we derive three insights, which have informed the design of New Snow:

- (1) We understand the deep generative models as reflecting qualities of the data from which it is trained.
- (2) Exploration of datasets and models can happen through non-verbal means.
- (3) Learning how to *prompt* a model effectively means building an understanding of the data on which it is trained.

3 CONCEPT AND INTERACTION

New Snow is a project that explores how exploring an image synthesis model using drawing actions as a means of *prompting* lets the user learn about certain qualities of the underlying data.

From the digital collection of MUNCH, we have identified 5800 uncolored, ink, or pencil drawings made by Edvard Munch (see examples in fig. 2). Based on these drawings we have trained a StyleGAN 2 model [14, 16] and then a pixel2style2pixel (pSp) model [22]. Together this allows us to map drawings made by a user into the model and synthesize sketches from this input.

The prototype consists of a table with a matte transparent surface. The user places a piece of tracing paper on the surface and draws with a pen (fig. 3). Underneath the table, a camera tracks the lines on the paper and sends them to the pixel2style2pixel model. From the lines on the paper, the model synthesizes an image based on Edvard Munch’s sketches (fig. 1). This image is projected back onto the tracing paper for the user to see. As the user draws or moves the paper around, the system continuously and multiple times per second updates the synthesized image to match. This

allows the user to explore the qualities of the model through an almost conversational relation to the system.



Figure 3: The top image is a render of the prototype with the sides open, so the inner structure can be seen. The image below shows a user interacting with the drawing interface.

The aim of this project is to help the user develop an attention to the aesthetic of Edvard Munch’s drawing practice. We are not attempting to explain the technical details of his practice nor the historical or biographical relations. However, through the embodied engagement with the visual aesthetic derived from his drawings, we expect the user to develop a sense of the visual qualities related to Munch’s sketching and drawing practice. That does not necessarily require the original works to be reproduced, the aim is rather to create a focus on the dynamics and patterns in Munch’s sketching practice that a person not skilled in the act of drawing or analyzing drawings might not otherwise have noticed.

Models like Dall-E, Midjourney, and Stable Diffusion have become famous for their ability to synthesize coherent images from almost any prompt in the style of well-known artists. In comparison, this model provides much more resistance. It does not draw for the user, but it responds to their drawings and attempts to expand and complete them. Due to the nature of the underlying model, the user will have to adapt a particular drawing strategy in order to achieve the greatest level of control, as the model does not respond to symbolic representation but rather the saliency of particular lines constituting an image.

This tension requires the user to explore the workings of the model to understand how it works and what it responds to. In that way, the function of the model is to provide a space to explore rather than being a tool to reach other ends. The exploration is by proxy an exploration of Munch’s sketching practice. It asks the user

to contemplate and then draw not *what* Munch might have drawn, but *how* he might have drawn.

At the core of the experience is the user interaction with the DGM, however, that interaction happens within an embodied and material context. People’s bodily relations to artworks shape how they might cast the art objects in specific cultural roles i.e. as a commodity, a fragile piece of history, or a toy [24]. With this awareness we expect the embodied relation to the drawings in New Snow to influence people’s cultural connotations of the drawing activity. Thus we have opted for a physical setup where the user performs the drawing action with an actual pen on paper. First, the paper and pen have other affordances than a touchscreen and pen interface, one being that erasing is not possible, and the drawn image can be moved around on the surface, lifted off the surface, and brought along. Secondly, the tactile feeling of pen and paper differs significantly from the glass surface of a tablet and e-pen and evokes different connotations and importantly a closer material connection to the tools used by Edvard Munch.

4 TECHNICAL DESCRIPTION

The drawing table is built into a flight case 110 cm tall with a semi-transparent polycarbonate window on top. The drawing surface is lit with infrared (IR) light by LEDs within the table to eliminate shadows. An IR-sensitive camera fitted with an 850nm filter records the drawing surface (fig. 4). This is to avoid interference from the visible light cast by the projector. With software made with TouchDesigner, OpenCV, and Python the video feed is pre-processed into binary images, isolating the lines drawn on the paper. This image is submitted to the pixel2style2pixel model and a synthesized drawing is returned within a second. Adjustments are made to saturation and contrast before the projected image interpolates from the current to the new image.

4.1 Machine Learning

DGMs have seen tremendous progress in recent years and Generative Adversarial Networks (GANs) [9] have become one of the most influential deep generative architectures. Recently, inspired by style transfer [7, 13], the StyleGAN family of models [14–16] have been shown to give state-of-the-art results across a wide variety of image generation tasks [4]. Due to the exceptional quality of the images generated by StyleGAN models, the architecture has been called one of the most intriguing and well-studied architectures in recent times [1].

After training, StyleGAN has learned a mathematical space, denoted the *latent* space, where each point in the space corresponds to a unique image. The GAN is trained such that the distribution of the generated images follows the distribution of the images in the training data, both with respect to image quality as well as the internal variation between images in the training data.

The latent space is smooth, which means that if we interpolate between two points in the latent space, e.g. two points corresponding to portraits, the corresponding generated images will change gradually from one portrait to the other where each intermediate image is itself fully self-consistent and resembles Munch’s style in its own right. Thus the latent space can be seen as a representation of the space between all Munch sketches in the data set.

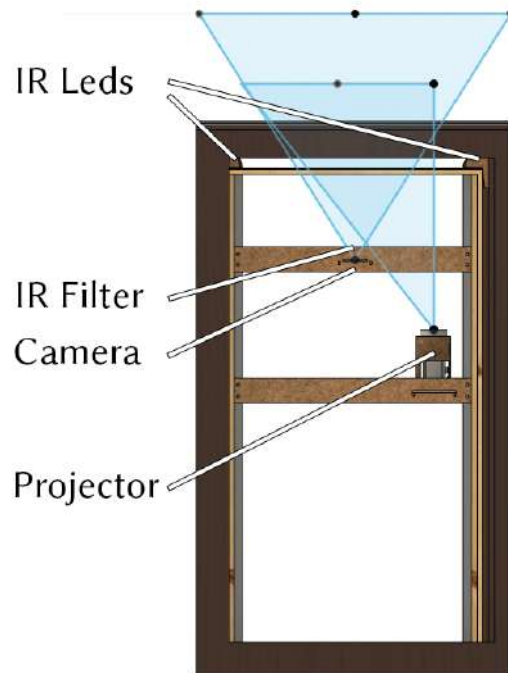


Figure 4: The prototype house a pico projector and a camera that are aligned with the drawing surface. An infrared filter removes the visible light from the camera input. Right below the top are two strips of infrared LED that illuminate the drawing surface. The lower part of the table can house a PC for processing the images.

To allow for direct user interaction with this latent space, we have trained a pSp encoder [22] which is able to map user-provided pen-and-paper sketches, into the latent space, thus transforming the user input to a sketch that follows the style of Munch.

5 AESTHETIC DRAWING STRATEGIES

In preliminary testing of the prototype, we have seen participants engaging actively with the drawing task. Participants apply widely different strategies, and we see indications that certain mental models yield more satisfying interactions than others. When users draw conceptually, e.g. a simplified house or tree, the system generates only limited visual response since these shapes lie far away from the images in the dataset. A more fruitful drawing strategy seems to be drawing one long stroke at first and then looking at the response of the system. Often a variety of more or less defined lines will appear. These lines can be reinforced or challenged by drawing other lines in the same area, which often results in more defined shapes and features, and the process can continue and evolve into a meaningful drawing. These are the strategies that we are interested in exploring and tuning the system to support.

6 CURATING THE DATASET

As the aesthetic qualities in Edvard Munch's sketching practice are mediated through the DGM, particular attention needs to be paid to the ways in which the sketches have been prepared for training and how the chosen model interprets the data. The images constituting the dataset for the StyleGAN model have been cropped from photographs of notebooks or loose paper sheets by human annotators that have made decisions on composition and the tightness of the crop to leave out damaged paper, smudges, handwritten notes, and other artifacts that have been deemed irrelevant for the project. This curation shapes the concept of the images created by the model. It determines what belongs to a drawing, and where on the page certain shapes will appear. Another limitation is the necessity for this type of model to be trained with square images. This requires the input images to be either stretched or cropped to fit this requirement. These issues reappear when the pSp model is trained as the input images are simplifications derived from the syntheses. However, the amount of simplification determines how far from the input image the synthesized images will be visually. This means that a significant part of the interaction design lies in the data curation process, making the iterative loop longer and more time-consuming than when designing heuristic interactive systems.

ACKNOWLEDGMENTS

The authors would like to thank MUNCH, Oslo for helping in developing the concept and funding the prototype. We are thankful to the research division at Random International for the development of the tracking system and AIRLab at IT University of Copenhagen for supporting the physical realization of the prototype.

REFERENCES

- [1] Yuval Alaluf, Or Patashnik, Zongze Wu, Asif Zamir, Eli Shechtman, Dani Lischinski, and Daniel Cohen-Or. 2022. Third Time's the Charm? Image and Video Editing with StyleGAN3. <http://arxiv.org/abs/2201.13433> arXiv: 2201.13433.
- [2] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–13. <https://doi.org/10.1145/3290605.3300233>
- [3] Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Benetot, Siham Tabik, Alberto Barbado, Salvador Garcia, Sergio Gil-Lopez, Daniel Molina, Richard Benjamins, Raja Chatila, and Francisco Herrera. 2020. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion* 58 (June 2020), 82–115. <https://doi.org/10.1016/j.inffus.2019.12.012>
- [4] Amit H. Bermano, Rinon Gal, Yuval Alaluf, Ron Mokady, Yotam Nitzan, Omer Tov, Or Patashnik, and Daniel Cohen-Or. 2022. State-of-the-Art in the Architecture, Methods and Applications of StyleGAN. <https://doi.org/10.48550/ARXIV.2202.14020>
- [5] Andreas Blattmann, Robin Rombach, Kaan Oktay, and Björn Ommer. 2022. Retrieval-Augmented Diffusion Models. <https://doi.org/10.48550/ARXIV.2204.11824>
- [6] Finale Doshi-Velez and Been Kim. 2017. Towards A Rigorous Science of Interpretable Machine Learning. <http://arxiv.org/abs/1702.08608> arXiv:1702.08608 [cs, stat].
- [7] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. 2015. A Neural Algorithm of Artistic Style. <http://dblp.uni-trier.de/db/journals/corr/corr1508.html#GatysEB15a>
- [8] Adarsh Ghosh and Devasenathipathy Kandasamy. 2020. Interpretable Artificial Intelligence: Why and When. *American Journal of Roentgenology* 214, 5 (May 2020), 1137–1138. <https://doi.org/10.2214/AJR.19.22145> Publisher: American Roentgen Ray Society.

- [9] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative Adversarial Nets. , 2672–2680 pages. <https://proceedings.neurips.cc/paper/2014/file/5ca3e9b122f61f8f06494c97b1afccf3-Paper.pdf>
- [10] Google Creative Lab, Bastien Girschig, and Romain Cazier. 2020. Draw to Art: Shape Edition. <https://experiments.withgoogle.com/draw-to-art-shape>
- [11] Google Creative Lab, Google Art & Culture Lab, and IYOIYO. 2018. Draw to Art. <https://experiments.withgoogle.com/draw-to-art>
- [12] Joana Hois, Dimitra Theofanou-Fuelbier, and Alischa Janine Junk. 2019. How to Achieve Explainability and Transparency in Human AI Interaction. In *HCI International 2019 - Posters (Communications in Computer and Information Science)*, Constantine Stephanidis (Ed.). Springer International Publishing, Cham, 177–183. https://doi.org/10.1007/978-3-030-23528-4_25
- [13] Xun Huang and Serge Belongie. 2017. Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization. <https://doi.org/10.48550/ARXIV.1703.06868>
- [14] Tero Karras, Miika Aittala, Janne Hellsten, Samuli Laine, Jaakko Lehtinen, and Timo Aila. 2020. Training Generative Adversarial Networks with Limited Data.
- [15] Tero Karras, Miika Aittala, Samuli Laine, Erik Härkönen, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2021. Alias-Free Generative Adversarial Networks.
- [16] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2020. Analyzing and Improving the Image Quality of StyleGAN.
- [17] Sarah Kenderdine and Heidi McKenzie. 2013. A war torn memory palace: Animating narratives of remembrance. In *2013 Digital Heritage International Congress (DigitalHeritage)*. IEEE, Marseille, France, 315–322. <https://doi.org/10.1109/DigitalHeritage.2013.6743755>
- [18] Sarah Irene Brutton Kenderdine and Tim Hart. 2011. Cultural Data Sculpting: Omni-spatial Visualization for Large Scale Heterogeneous Datasets. In *Museums and the Web 2011: Proceedings*, J. Trant and D. Bearman (Eds.). Archives & Museum Informatics, Toronto. http://conference.archimuse.com/mw2011/papers/cultural_data_sculpting_omnispatial_visualization_large_scale_heterogeneous_datasets
- [19] Bernd Lintermann. 2012. Beyond Cinema. Hongik University, Korea. https://www.bernd-lintermann.de/papers/Beyond_Cinema_Lintermann.pdf
- [20] Jonas Oppenlaender. 2022. A Taxonomy of Prompt Modifiers for Text-To-Image Generation. <http://arxiv.org/abs/2204.13988> arXiv:2204.13988 [cs].
- [21] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical Text-Conditional Image Generation with CLIP Latents. <https://doi.org/10.48550/ARXIV.2204.06125>
- [22] Elad Richardson, Yuval Alaluf, Or Patashnik, Yotam Nitzan, Yaniv Azar, Stav Shapiro, and Daniel Cohen-Or. 2021. Encoding in Style: a StyleGAN Encoder for Image-to-Image Translation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Nashville, TN, USA, 2287–2296. <https://doi.org/10.1109/CVPR46437.2021.00232>
- [23] Jeffrey Shaw, Neil Brown, Dennis Del Favero, Matt McGinity, and Peter Weibel. 2006. T_Visionarium II. https://www.jeffreyshawcompendium.com/portfolio/t_visionarium-ii/
- [24] Christian Sivertsen and Anders Sundnes Løvlie. 2021. Handling digital reproductions of artworks. *Journal of Somaesthetics* 7, 2 (2021), 21.